CSEE 4119 Computer Networks Midterm (Spring 2026)

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1. (15 pts) We learned that the DNS protocol primarily uses UDP, and the HTTP protocol uses TCP.
(1) (4 pts) When 20 hosts are concurrently sending DNS queries to a DNS server, how many socket connections does the DNS server maintain?
(2) (4 pts) When 20 clients are simultaneously connecting to a web server, how many socket connections does the web server maintain?
(3) (7 pts) A web client is requesting a web page (index.html) from a web server using HTTP 1.1 w/ persistent connections and pipelining. The index.html file has four embedded jpeg images and one style file (style.css). The style.css file contains two embedded jpeg images. Assume each image and file can fit into one packet. How many RTTs are needed to receive the webpage fully?

(1). DNS use UDP, so sockets number is 1.

(2). web server may use http, which relies on TCP → so socket number is 20.

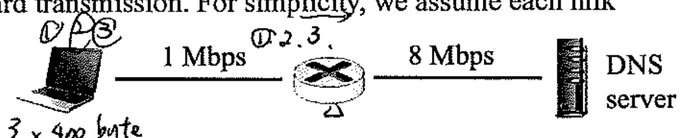
(3) client → server   tcp establish + html request

$2 + 1 = 3$ RTTs

- index.html
  - jpg1
  - jpg2
  - jpg3
  - jpg4
  - css
    - jpg5
    - jpg6.

⇒ are send in one time due to pipelining mechanism.

2. (15 pts) Suppose your laptop sends 3 DNS queries to a local DNS server via a router in between. The link rate between your laptop to the router is 1 Mbps, and the link between the router to the DNS server is 8 Mbps. Once receiving a query, the DNS server responds immediately with a response back to your laptop. Each DNS query results into a 400-byte physical frame, while each DNS response is a 500-byte physical frame. The router uses store-and-forward transmission. For simplicity, we assume each link allows full-duplex transmission and ignore any propagation delay, queuing or processing delay at the router. The laptop does NOT wait for the DNS response of the previous query before sending the next query.



(1) (2 pts) How long does it take for the DNS server to receive the first query?
(2) (5 pts) What is the time between your laptop sends the first bit of the first query and receives the last bit of the first DNS response?
(3) (8 pts) How long would it take for your laptop to receive all three DNS responses?
Give all answers in milliseconds.

(1). first query reaches DNS: $\frac{400\times 8 \text{ bits}}{1\times 10^6 \text{ bps}} + \frac{400\times 8 \text{ bits}}{8\times 10^6 \text{ bps}} = 3.2\times 10^{-3}s + 4\times 10^{-4}s = 3.6\times 10^{-3}s = \cancel{0.36ms}$ 3.6ms

(2) time line: $0 \longrightarrow 3.2\times 10^{-3} \longrightarrow 3.6\times 10^{-3} \longrightarrow$
begin   first query router   first query DNS

for the DNS server back to laptop: $t_2 = \frac{500\times 8}{8\times 10^6} + \frac{500\times 8}{1\times 10^6} = 5\times 10^{-4}s + 4\times 10^{-3} = 4.5\times 10^{-3}s = \cancel{0.045ms}$ 4.5ms

$t_{total} = 3.6\times 10^{-3}s + 4.5\times 10^{-3}s = 8.1\times 10^{-3}s = 8.1ms$

(3). $t_3 total = t_{total} +$ ~~caused~~ because bottleneck is the ~~first~~ host to the router.

$= 8.1ms$

wireless

hostA host B

wire

3. (10 pts) We are implementing a reliable transport protocol for two hosts to share data reliably. The two hosts are connected by a wired channel and a wireless channel. The wireless channel is quite lossy while the wired channel has a much lower loss rate. These two channels are used to send data and acknowledgements separately to ensure that data and control channels are orthogonal and separated.

(1) (5 pts) If we use the wireless channel to send data and the wired channel to send acknowledgements, and we must choose between Go-Back-N and Selective Repeat, which protocol should we choose and why?

(2) (5 pts) If we use the wireless channel for acknowledgement and the wired channel for data, should we choose Go-Back-N or Selective Repeat? Why?

(1). wireless wired

Sender receiver

↓ ↑

more loss more secure

we should focus on packet loss

→ Go-Back-N ensures retransmission from sender of any loss packet

(2) We focus more on Ack loss

sender receiver

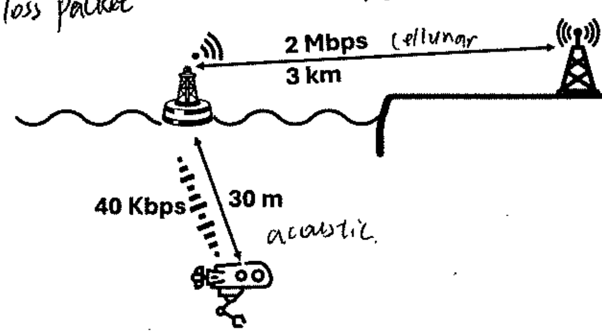
↓ ↓

secure more loss

Selective Repeat introduce buffer at receiver.

→ Selective Repeat. ensures a buffer in receiver, so that it will send out less ACKs than Gobach-N, therefore less loss

4. (30 pts) Let us consider a networking system for underwater robots to send data to a ground base station. The robot communicates with a buoy node on the water surface via an acoustic communication link, and the buoy node then transmits the data to the ground base station via a cellular communication link. The right figure depicts the distances and link bandwidth numbers for each link/hop.



(1) (4pts) What is the propagation delay for the underwater link (i.e., from the robot to the buoy), and the air link (from the buoy to the base station)? The speed of sound underwater is 1.5 km/s, and the speed of light is 3×10^8 m/s.

(2) (5pts) If the robot is sending a frame of 2 Kbits in length to the ground station, how long does it take for the ground station to receive the full frame?

0.07101s

(1). $d_{under} = \dfrac{30m}{1.5\times10^3 m/s} = 2\times10^{-2}s$

$d_{air} = \dfrac{3\times10^3 m}{3\times10^8 m/s} = 1\times10^{-5}s$

(2). $d = d_{prop_1} + d_{prop_2} + d_{trans_1} + d_{trans_2}$.

$= 2\times10^{-2} + 1\times10^{-5} + \dfrac{2\times10^3}{40\times10^3} + \dfrac{2\times10^3}{2\times10^6}$

$= 2\times10^{-2} + 1\times10^{-5} + 0.05 + 10^{-3} = 0.02 + 0.00001 + 0.05 + 0.001$

(3) (7pts) The robot sends to the ground station an image that is split into 100 frames each with 2Kbits, how long does it take for the ground station to receive the whole image?

(4) (4pts) During the time when the robot is continuously transmitting each of the 100 frames, what is the utilization ratio of the cellular link (i.e., the link between the buoy and the ground station)? Let's assume that there are no other users using this cellular link.

(5) (10pts) Now we consider the transport protocol at the robot and the base station. We will let the base station send an ACK to the robot upon receiving a frame, where each frame is 2 Kbits. What is the acoustic link utilization when we use the Stop and Wait protocol (i.e., sliding window of one packet/frame)? What if the frame length is 200 bits? Is Stop and Wait a good choice in the latter case and why?

(3) $d_3 = d_{prop1} + d_{prop2} + d_{trans1} + d_{trans2}$

$$= 0.02 + 0.00001 + \frac{2\times10^3\times100}{\cancel{4\times10^4}\ 40\times10^3} + \frac{2\times10^3\times100}{2\times10^6}$$

$$= 0.02+0.00001 + \frac{2\times10^5}{4\times10^4} + \frac{2\times10^5}{2\times10^6}$$

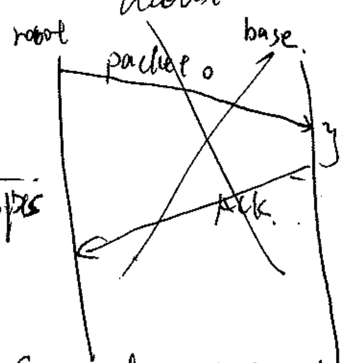
$$= 0.02+0.00001+5+0.1$$

$$= 5.12001\ s$$

Sorry, I don't have a calculator.

(4) $$U = \frac{\text{arrival rate of bits}}{R} = \frac{\frac{100\times 2\text{Kbits}}{d_3}}{2\text{Mbps}} = \frac{\frac{2\times10^5}{5.12001}}{2\times10^6} = \frac{\frac{1}{5.12001}}{10} = \ldots$$

(5)



$U = \frac{R_{real}}{40\times10^3\ bps}$ R

① when one frame is 2×10^3 bits, $\frac{L}{R}$ = acoustic transmission time $= 0.05s$

$$U = \frac{\frac{L}{R}}{RTT + \frac{L}{R}} = \frac{0.05}{0.05 + RTT} = \frac{0.05}{0.05+0.00001\times2} = \frac{\frac{2000}{t}}{4\times10^4 bps}$$ where $t = (2\times10^{-2}+1\times10^{-5})\times2 + 0.05 + 0.001$

② when frame length is 200 bits

$\frac{L}{R}$ = transmission time $= \frac{200}{40\times10^3}$

$$= \frac{\frac{200}{t_{total}}}{4\times10^4 bps}$$ where $t_{total} = (0.02+10^{-5})\times2 + \frac{200}{4\times10^4} + \frac{200}{2\times10^6}$

No. Stop and wait has low utilization, we should use pipeline to reduce waste.

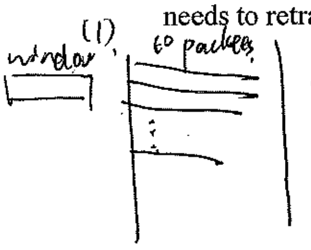
5. (15pts) Suppose you have an Internet-of-things (IoT) device with limited memory. Your computer sends data to the IoT device via a selective-repeat protocol. They will also have an initial handshake process to determine sliding window size, initial sequence number, etc. Your computer supports a sliding window of 4095 packets. The IoT device, however, can only buffer at most 4 packets due to its memory constraint. The round-trip time (RTT) is large enough to allow for 10 packets in transit at a given point in time (5 in each direction).

(1) (7 pts) The IoT device believes it can handle 10 packets and advertises a window size of 10. Can you describe what will happen in this case?

(2) (4 pts) For a batch of 10 packets, if only a single packet fails to arrive, is it possible that the sender needs to retransmit more than 6 packets? Why?

(2). No. the buffer only holds four packets' information, So it can at most send 4 Acks so the sender will at most re-transmit four.

(1)



buffer size = 4.

$\boxed{0123}$ 4 5

If any packet is lost among the ten packets but fall out side the receiver's buffer, it will not notice and send ACKs. If packets arrive but fall out of the buffer, it will not send ACKs as expected.

(3) (4 pts) How should the IoT device advertise the window size during the initial handshake?

(3). It should send out One packet at first and graduallay grow up the number. When the window size is too large ~~enough~~ ~~so~~ there the send have to retransmit, the window should shrink.

By sending out ~~the window~~ size sequences number ~~indicating~~ as X, X + windowsize, ~~X+~~ ...... $X + n$ windowsize

6. (15 pts) Amazon provides video streaming for popular movies. Suppose a movie is 1 GB in size. The number of users that are interested in the movie is N = 100. The users have mixed upload/download capacity. Specifically, 60% of users have 10Mbps upload and 20Mbps download speed, and 40% of users have 5Mbps upload and 10Mbps download speed. The server upload rate is 40Mbps.

(1) (6 pts) Amazon uses a client-server architecture. What is the minimal time to distribute the movie to all users?

(2) (7 pts) If Amazon uses a BitTorrent style peer-to-peer distribution system, what is the minimal time to distribute the movie to all users? Assuming perfect distribution. No delay in peer matching.

(3) (2 pts) For the peer-to-peer distribution starts to outperform client-server distribution, what is the required N value? Why?

(1). $\frac{NF}{us} = \frac{100 \times 1GB}{40Mbps} = \frac{10^2 \times 10^9 \times 8}{40 \times 10^6} = \frac{8 \times 10^{11}}{4 \times 10^7} = 2 \times 10^4 s$

$\frac{F}{d_{min}} = \frac{1GB}{10Mbps} = \frac{8 \times 10^9}{10^7} = 800s$

$D_{cs} \geq \max\{2 \times 10^4 s, 800s\} = 2 \times 10^4 s$

(2). p-p. $\frac{F}{us} = \frac{1GB}{40Mbps} = \frac{8 \times 10^9}{4 \times 10^7} = 2 \times 10^2 s$

$\frac{F}{d_{min}} = \frac{8 \times 10^9}{10^7} = 800s$

$\frac{NF}{us + \Sigma u_i} = \frac{100 \times 8 \times 10^9}{40 \times 10^6 + 60 \times 10 \times 10^6 + 40 \times 5 \times 10^6}$

$= \frac{8 \times 10^4}{4 + 60 + 20} = \frac{8}{84} \times 10^4 s$

$D_{p\text{-}p} \geq \max\{200s, 800s, \frac{8}{84} \times 10^4 s\} = \frac{8}{84} \times 10^4 s$

(3). $D_{c\text{-}s} = \max\left\{\frac{N \cdot 1GB}{40Mbps}, \frac{1GB}{10Mbps}\right\}$

$D_{p\text{-}p} = \max\left\{\frac{1GB}{40Mbps}, \frac{1GB}{10Mbps}, \frac{100GB}{us + \Sigma u_i}\right\}$

$= \max\left\{\frac{1GB}{40Mbps}, \frac{1GB}{10Mbps}, \frac{100GB}{40 + 0.6N \cdot 10Mbps + 0.4N \cdot 5Mbps}\right\}$

bottle neck indentify

$D_{c\text{-}s} \geq D_{p\text{-}p}$

$\frac{N \cdot 1GB}{40Mbps} \geq \frac{100GB}{40Mbps + 0.6N \cdot 10Mbps + 0.4N \cdot 5Mbps} = \frac{8 \times 10^{11}}{4 \times 10^7 + 6 \times 10^8 + 2 \times 10^8}$

$\frac{N}{40} \geq \frac{100}{40 + 6N + 2N}$

$\geq \frac{100}{40 + 8N}$

Solve N